



DATAQUEST 3.0

ROUND 1



PROJECT DETAILS

PROJECT TITLE

**PRISM Engine: Multi-Dimensional STEAM
Career Guidance & Hyper-Local Innovation
Platform**

ONE-LINE PITCH

PRISM Engine converts student psychometrics, parental financial limits and live job-market signals into one scored, financially viable career roadmap that students and parents can both see.

TRACK

DQNM



TEAM & MEMBERS

TEAM

[Team Name]

PARTICIPANTS

- [Member 1 Name, Reg. No.]
 - [Member 2 Name, Reg. No.]
 - [Member 3 Name, Reg. No.]
 - [Member 4 Name, Reg. No.]
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02 Problem Definition

Who is impacted · Today's workarounds · Why current methods fail · Concrete proof

01 · AUDIENCE

WHO & HOW OFTEN

Students in schools and colleges across India, their parents and their educators. Parents are the primary financial and emotional decision-makers.

Crucial career decisions are made frequently, and over 90% of students have no structured, evidence-based guidance when they make them.

02 · STATUS QUO

CURRENT APPROACH

Students choose careers without knowing where professions are most in demand, what real salary trajectories look like, which interdisciplinary pathways exist or which scholarships they can access.

Parents are routinely left out of the analysis. Families default to rigid, conventional career choices.

03 · DEFICIENCY

WHY IT FALLS SHORT

- 1. Geographic demand opacity:** students do not know which regions and global markets value their skills most.
- 2. Economic yield mismatch:** no clarity on compensation trajectories or ROI after a course.
- 3. Parental exclusion and financial blind spots:** household affordability is never analyzed, leading to debt or misaligned expectations.
- 4. Unexplored career alternatives:** emerging hyper-local and STEAM innovation pathways are overlooked.
- 5. Disconnected aid discovery:** entrance exams, rankings and targeted scholarships are not visible in time.

04 · VALIDATION

EVIDENCE

Over 90% of Indian school and college students lack structured, evidence-based career guidance.

Source: DataQuest 3.0 DQNM problem statement.

Reported consequences: high career dropout rates, unfulfilled aspirations, graduate underemployment, severe family conflict and financially unsustainable career commitments.

03 Proposed Solution

System identity · End-to-end user workflow · Competitive advantages

SOLUTION CORE

PRISM Engine

A multi-dimensional career guidance and student SWOT analytics platform. It vectorizes student competencies and parental constraints, solves them against household financial limits, and blends the result with dynamic industry hiring indices to output clear, financially viable career pathways.

USER FLOW

1. USER INPUT

Students: psychometric responses on aptitude, cognitive preferences and interests.
Parents: affordability, risk appetite, socio-economic mobility and aspirations.



2. PROCESSING

Inputs normalized into vectors. Financial Constraint Solver filters by affordability; Parent-Student Conflict Index measures disagreement. Blended with hiring indices: job velocity, economic disruption, geographic demand, hyper-local STEAM.



3. OUTPUT

Composite scores, suitable career domains with alignment gaps, student SWOT analytics, and a roadmap of educational pathways, entrance exams, scholarships and development options.

ADVANTAGE

WHY IT IS BETTER

Most guidance looks at the student alone. PRISM Engine evaluates the student and the family together, so every recommendation already respects household affordability and shows where parent and student expectations differ. Recommendations follow current demand and salary outlook by region.

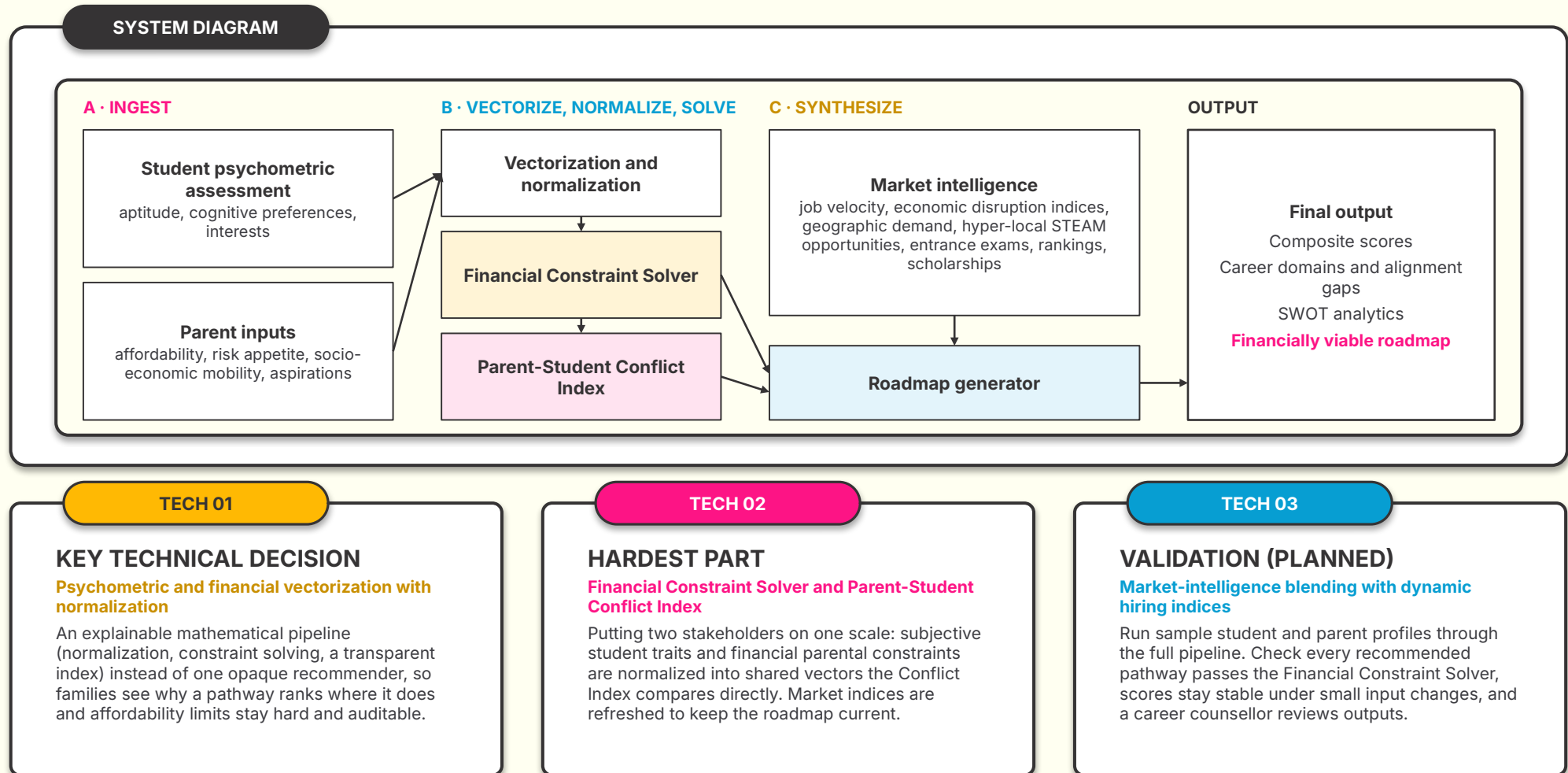
INNOVATION

WHAT IS DIFFERENT

Two explicit constraints sit at the core of the model:
Financial Constraint Solver: screens pathways against household affordability and risk appetite.
Parent-Student Conflict Index: quantifies disagreement before a family commits.
Hyper-local STEAM innovation pathways are scored alongside conventional careers.

04 System Architecture

System diagram is the primary visual · Engineering decisions · Validation



05 Feasibility & Scope

Concrete boundaries · Operational run rate · Risk mitigation · Post-hackathon plans

01 · BOUNDARIES

1. SCOPE

In: End-to-end guidance for one student profile: student and parent input forms, normalization, Financial Constraint Solver, Parent-Student Conflict Index, composite scores, career domain matches with alignment gaps, and a roadmap of pathways, exams and scholarships.

Out: Live national data feeds at scale, school-wide hyper-local deployments and counsellor dashboards for many students, all deferred beyond Round 1.

02 · ECONOMICS

2. RUNNING COST

The core engine is mathematical, so each assessment needs little compute. Main cost drivers are hosting, a database for profiles and market indices, and market-data access.

The prototype is designed for low-tier cloud resources.

Estimated monthly spend: [Monthly cost estimate in INR]

03 · CONTINGENCY

3. RISKS & FALLBACK

Stale or missing market data: fall back to the last verified index snapshot and show its date.

Incomplete or inaccurate parent inputs: use ranges and conservative defaults, and flag lower confidence.

Biased psychometric signals: combine several indicators before scoring and display the confidence level.

04 · HORIZON

4. AFTER THE HACKATHON

Next: live job-velocity, disruption-index and geographic-demand feeds, plus a maintained database of entrance exams, institutional rankings and scholarships.

Then: production hardening with consent and privacy controls for student and family financial data.

Later: school pilots and hyper-local STEAM innovation pathways, in line with Nano Mech Labs' goal of marketable solutions from school ecosystems.

06 Impact & Metrics

Primary measurable impact · Transparent calculation · Stakeholder advantages

PRIMARY METRIC

90%+

of Indian school and college students lack structured, evidence-based career guidance today.

This is the gap PRISM Engine targets.

HOW IT WAS MEASURED

The 90% figure comes from the DataQuest 3.0 DQNM problem statement and serves as the baseline. PRISM Engine's own impact will be tracked with three planned measures:

1. Share of recommended pathways that pass the Financial Constraint Solver
2. Change in the Parent-Student Conflict Index after guidance
3. Number of alignment gaps closed

These are targets to confirm in pilots.

IMPACT 01

WHAT IMPROVES

Before: choices made without demand data, salary outlook, scholarship visibility or the parents' input.

After: one composite-scored, financially viable roadmap, with alignment gaps and parent-student conflict visible before the family commits.

IMPACT 02

WHO ELSE BENEFITS

Parents gain a clear view of affordability and risk.

Educators and schools gain student SWOT analytics.

Nano Mech Labs' school ecosystems and local industry gain hyper-local, marketable STEAM solutions.

IMPACT 03

RISKS / MISUSE

Risks: over-reliance on a single score, biased psychometric signals, exposure of family financial data, outdated market data.

Mitigation: explained scores as guidance for counsellor-supported decisions, several indicators with confidence levels, consent and minimal data retention, dated market snapshots.

07 Why This Project?

Direct contrast with today's alternatives · Unfair technical advantage

STATUS QUO

CURRENT APPROACH

Career guidance today is generic and student-only. Parents are left out of the analysis, and there is little visibility into regional demand, salary trajectories, ROI, entrance exams, rankings or scholarships.

Families end up with rigid, conventional choices, debt or conflict.

OUR SOLUTION

OUR APPROACH

PRISM Engine takes in student psychometrics and parental financial parameters, normalizes them, applies the Financial Constraint Solver and the Parent-Student Conflict Index, and blends the result with dynamic hiring indices.

The output is a market-aligned, financially viable roadmap that includes hyper-local STEAM pathways.

MOAT / DIFFERENTIATION

KEY DIFFERENCE

PRISM Engine treats the family as part of the model: a Financial Constraint Solver and a Parent-Student Conflict Index decide which pathways are viable before any career is recommended.

08 Extra Slide: Problem Statement Coverage

Optional appendix · Extra diagrams, product screenshots, benchmarks, or backup details

REQUIREMENT (DQNM problem statement)	HOW PRISM ENGINE COVERS IT
1. Assess student aptitude, cognitive preferences and interest indicators	Psychometric responses are vectorized and normalized.
2. Consider financial constraints, risk appetite and family considerations	Parent inputs feed the Financial Constraint Solver.
3. Map profile factors against evolving sector demand and career options	Normalized vectors are blended with hiring indices: job velocity, economic disruption, geographic demand.
4. Combine student metrics and family parameters into composite scores	Composite scoring plus the Parent-Student Conflict Index.
5. Identify suitable career domains and highlight alignment gaps	Domain matching with gap flags and SWOT analytics.
6. Recommend educational pathways, examinations and development options	A roadmap listing pathways, entrance exams, scholarships and development options.
7. Close the five problem gaps	Geographic demand view, salary and ROI view, parent inputs, hyper-local STEAM pathways, exam and scholarship discovery.